

PRATYUSH PRAKHAR

Delhi, India | +91-7042236696 | pratyushprakh@gmail.com

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PROFESSIONAL SUMMARY

Cybersecurity Engineer, 4+ years across AppSec, cloud security, and GenAI security. Three of those years on the AWS side (direct at Amazon and through AWS-focused consulting), working on 15+ high-risk services, fixing 150+ critical vulnerabilities, and cutting attack surface up to 30% with STRIDE threat modeling and zero-trust controls. Worked on Amazon Bedrock from its early days to help build the GenAI security baseline: OWASP LLM Top 10 controls, prompt injection guardrails, adversarial ML defenses. Pursuing OSCP. Now bringing the same security and AI rigor to private and pre-IPO market research at Bridging Gap.

TECHNICAL SKILLS

Application Security: SAST/DAST, API Security, STRIDE/PASTA Threat Modeling, OWASP Top 10, OWASP LLM Top 10, Secure Code Review, Vulnerability Management, CVSS, CVE Management, SSDLC, AI Red Teaming, Zero-Trust Architecture

Cloud & Infrastructure: AWS (IAM, S3, EC2, GuardDuty, Security Hub, CloudTrail, KMS, Amazon Inspector, AWS Config), CI/CD Security, SIEM/SOAR Implementation, Secrets/Token Management, PCI-DSS, NIST CSF, ISO 27001, SOC 2

AI/GenAI Security: LLM Security, Prompt Injection Defense, Adversarial ML Defense, Memory Poisoning Defense, Tool Misuse Controls, RAG Security, Output Validation/Sanitization, Model Risk Assessment, AI Supply Chain Security, NIST AI RMF, AWS Bedrock Security, LangChain/LangGraph, CrewAI, PydanticAI, Pi Agent, Claude Code, Codex

Offensive Security: Penetration Testing (OSCP methodology), API Penetration Testing, Burp Suite, Metasploit, Nmap, Nessus, Wireshark, SQLMap, Hydra, MITRE ATT&CK Mapping

Languages & Databases: Python, Golang, TypeScript, JavaScript, Bash/Shell, SQL, HTML/CSS, MySQL, DynamoDB

Regulated Domain Exposure: Financial Data Governance, Audit Reports, Private/Unlisted Markets, RBI Cybersecurity Framework, DPDP Act, EU AI Act

CERTIFICATIONS

- AI Security Fundamentals | Microsoft Learn
- Security for AI Essentials | IBM SkillsBuild
- AI Fluency: Framework & Foundations | Anthropic Academy
- OWASP Top 10, OWASP LLM Top 10 | PortSwigger
- Web Pentesting, Burp Suite, Metasploit | TryHackMe
- OSCP Preparation | Offensive Security methodology | Expected Dec 2026

PROFESSIONAL EXPERIENCE

AI Research Engineer, Private Markets | Bridging Gap Investment Services, Mumbai

October 2025 - Present

- Building agentic systems on LangGraph, Pi Agent, OpenAI Codex, and Claude Code to revamp the largely manual workflows that run pre-IPO and private market operations: company audits, financial review, tax positioning, regulatory filings, and management screening. Cut review time on a typical company from multiple days to under a day, freeing analysts to spend that time on judgment work instead of data assembly.
- Built the security and data governance layer behind the private-market data pipelines: scoped access controls, encryption at rest and in transit, audit logging, and PII boundaries on confidential filings, audit reports, and regulator data. Aligned the controls with RBI cybersecurity guidance, DPDP, and EU AI Act expectations so the AI tooling can scale across analyst teams without breaking compliance posture, and removed the need for analysts to manually anonymize source documents on every run.
- Restructuring the firm's financial models so each step of an instrument analysis pipeline (valuations, comparables, screening logic, sector cuts) can run on grounded data from filings, audit reports, and analyst notes. The goal is measurable lift on speed and efficiency without giving up correctness or recall, with the analyst still in the loop on the final read.

Sr. Security Analyst | Solgenie Technologies Inc., USA

January 2024 - December 2024

- Carried the AWS security work into consulting for financial services and SaaS clients, surfacing critical issues, prioritizing fixes by business risk, and trimming cross-account AWS attack surface by 20% through tighter API and IAM controls.
- Ran AppSec consulting work for clients across vulnerability assessments, secure code reviews, and OWASP Top 10 reviews on APIs and microservices, and cut assessment turnaround by 40%, saving roughly \$10K/month in their delivery costs.
- Wrote a Generative AI layer over SAST triage and incident response, lifting triage accuracy ~10% and shaving days off the remediation loop in AWS-heavy environments.
- Set up a standard pentest playbook across 4+ client engagements using Burp Suite, Metasploit, and Wireshark, so each new project started from the same baseline instead of getting redesigned every time.

Application Security Engineer | Amazon Web Services, New York, USA

July 2021 - December 2023

- Held the security design review seat for high-risk AWS services including CodeSuite, IAM Console, and the SDKs. Pushed global security standards across 15+ services and walked the teams through NIST CSF, ISO 27001, PCI-DSS, and SOC 2 review cycles, doing the vulnerability triage and architectural control design alongside the service teams.
- Led AppSec on Amazon Bedrock from preview through GA. Designed the threat models and put the early security controls in place (memory poisoning defenses, tool misuse controls, prompt injection guardrails) for AI infrastructure that enterprise customers in regulated industries depended on. Built adversarial ML defenses against OWASP LLM Top 10 and brought open high-risk findings down 25% over four review cycles.
- Ran whitebox, greybox, and blackbox assessments across AWS services, looking for architectural flaws as much as code-level bugs. Pushed SAST into the SDLC pipelines, where it caught 150+ critical vulnerabilities before code shipped.
- Used STRIDE and MITRE ATT&CK to drive proactive threat hunts on AWS workloads. Tracked 10-15% security posture improvements at service level over the engagements, mostly from removing dormant attack paths and tightening high-risk surface.
- Wrote Python-based SIEM tooling and folded OS patching into CI/CD pipelines. Cut manual touch-points on deploys and brought incident response time down by about 30%.
- Wrote the service-level security docs and ran cross-team review sessions where product and engineering teams brought questions they did not know who else to ask. Set security standards for the org, mentored newer engineers on secure SDLC, and kept security posture consistent across multiple AWS service teams instead of letting it drift per team.

Application Security Intern | Varian Medical Systems, Palo Alto, USA

June 2020 - September 2020

- Reviewed application access security on SAP, HR-Workday, and ServiceNow, and rebuilt the RBAC layer across all three. Brought unauthorized-access incidents down 20% across a 10,000+ employee base, with controls aligned to standard enterprise access management practice.
- Hardened the SAP S/4HANA authorization model, cleaned up PFCG roles and audit controls in the CRM, and tightened privilege boundaries. Cut access-vulnerability surface by 15% on the RBAC review pass alone.
- Built a ServiceNow ticketing automation prototype that cut about 6 hours/week of duplicate work for the team and tested out secure-by-default patterns for the enterprise application crews.

Security Analyst | SPRITZ Security & Privacy Group, Università di Padova, Italy

June 2018 - September 2018

- Built an IDS for Smart Grid networks targeting Sybil and Wormhole attacks on RPL-based Industrial IoT mesh. Hit 98% detection accuracy using the Highest Rank Common Ancestor verification principle, with a custom wormhole detection algorithm running inside the CONTIKI-OS COOJA simulator against modelled Wireless Sensor Network attack scenarios.

RESEARCH & OPEN SOURCE CONTRIBUTIONS

AI Agent Research & Development

Ongoing

- Asobiba is the main project: an open-source factory layer that sits over AutoGen, PydanticAI, SmolAgents, OpenAI Agents SDK, and LlamaIndex, with a security tooling layer on top, designed for AI-assisted offensive security workflows.
- F.R.I.D.A.Y. and pi-friday push the same ideas into agentic research systems, layering in web intelligence, semantic caching, and multi-agent orchestration.

Financial Intelligence & Software Development

Ongoing

- Kinyu is the financial data and AI research toolkit I am building for listed and unlisted market work. It pulls fragmented company filings, audit reports, and management commentary into a single layer that is actually queryable for an analyst.
- Chishikiko sits behind it as a personal knowledge base on Python, Golang, ML, and systems programming, written as I learn and pulled from when building the production tooling.

Security Research & CTF

Ongoing

- Working through HackTheBox and TryHackMe boxes on the side, mixing the standard pentesting toolset (Nmap, Metasploit, Burp Suite, SQLMap) with companion AI tooling I am building to automate the enumeration and early exploit development steps.
- Writing up the boxes I solve so the methodology stays useful for the next person, and so I can find it again later.

EDUCATION

M.S., Project Management - Cyber Security Operations, Harrisburg University Coursework: Agile/Scrum Methodology, Securing App and Software, Cloud and Network Security	August 2024 - July 2025 GPA: 3.8/4.00
M.S., Cyber Security Engineering, University of Southern California (USC) Coursework: Web Application Security, Security Assurance & Privacy, Cryptography	August 2019 - May 2021 GPA: 3.85/4.00
B.Tech., Electronics & Communication Engineering, Delhi Technological University Coursework: Computer Networks, Operating Systems, Algorithms & Data Structures	August 2015 - May 2019 GPA: 8.22/10.00